

Test 1 Revision Paper

VCE Algorithmics (HESS) · Unit 4 AOS1 · Criterion 5 practice

Same structure and mark allocation as Friday's test. Time yourself: 45 minutes.

Worked solutions on the last page — attempt everything first.

Question 1 — Growth classes

(3 marks)

Order the following from slowest-growing to fastest-growing:

$n!$, n^2 , $\log n$, $n \log n$, 2^n , n

Question 2 — Tightest form

(4 marks)

Simplify each to tightest Big-O form (1 mark each):

a.) $O(9n + 500)$

b.) $O(n^3 + 50n^2 + n \log n)$

c.) $O\left(\frac{n(n+1)}{2} + 3n\right)$

d.) $O(3^n + n^2)$

Question 3 — The formal definition

(4 marks)

Using the formal definition of Big-O, show that $f(n) = 5n + 30$ is $O(n)$. State your constants c and n_0 explicitly and show the bounding inequality holds.

Question 4 — Analysing pseudocode

(12 marks)

For each algorithm, state the worst-case time complexity in tightest Big-O form and justify it in one or two sentences.

a.)

(3 marks)

```
1 Algorithm SumEvens(L):  
2   total ← 0  
3   Foreach x in L Do  
4     If x mod 2 = 0 Do  
5       total ← total + x  
6   Return total
```

b.)

(3 marks)

```
1 Algorithm PrintGrid(n):  
2   Foreach i from 1 To n Do  
3     Foreach j from 1 To i Do  
4       print(i, j)
```

c.)

(3 marks)

```
1 Algorithm Zoom(n):  
2   i ← 1  
3   While i < n Do  
4     i ← i * 4  
5   Return i
```

d.)

(3 marks)

```
1 Algorithm DoublePass(L):  
2   Foreach i from 1 To n Do  
3     j ← n  
4     While j > 1 Do  
5       print(L[i], j)  
6       j ← j div 2  
7   Foreach x in L Do  
8     print(x)
```

Question 5 — Best and worst case

(3 marks)

```
1 Algorithm ContainsTarget(L, t):  
2   Foreach i from 1 To n Do  
3     If L[i] = t Do  
4       Return True  
5   Return False
```

Describe a best-case input and a worst-case input, and give the time complexity of each using Θ notation.

Question 6 — Graph algorithms

(4 marks)

A delivery network is modelled with $|V|$ depots and $|E|$ roads, stored as adjacency lists.

a.) DFS is run to check whether all depots are reachable. State the tight worst-case bound and justify it by counting node-visits and edge-touches over the whole run. (2 marks)

b.) The network is then upgraded so that every depot has a direct road to every other depot. Rewrite your bound from part (a) in terms of $|V|$ only, showing the substitution. (2 marks)

Worked solutions

Q1. $\log n$, n , $n \log n$, n^2 , 2^n , $n!$

Q2. a) $O(n)$ b) $O(n^3)$ c) $O(n^2)$ — the sum expands to $\frac{n^2}{2} + \frac{n}{2} + 3n$; keep the dominant term d) $O(3^n)$ — exponentials dominate polynomials.

Q3. Take $c = 35$, $n_0 = 1$. For all $n \geq 1$: $30 \leq 30n$, so $f(n) = 5n + 30 \leq 5n + 30n = 35n = c \cdot n$. Hence $f(n) = O(n)$. (Any valid pair works, e.g. $c = 6$, $n_0 = 30$: for $n \geq 30$, $30 \leq n$, so $5n + 30 \leq 6n$.)

Q4a. $O(n)$ — a single loop of n iterations with an $O(1)$ body; the **If** does not add a factor.

Q4b. $O(n^2)$ — triangular loop: $1 + 2 + \dots + n = \frac{n(n+1)}{2}$ iterations; the constant $\frac{1}{2}$ and lower-order term drop.

Q4c. $O(\log n)$ — i multiplies by 4 each pass, so the loop runs $\lceil \log_4 n \rceil$ times; the base of the logarithm is a constant factor.

Q4d. $O(n \log n)$ — block 1: n outer iterations $\times \log_2 n$ inner (halving) $= O(n \log n)$; block 2: $O(n)$; sequence adds, $n \log n$ dominates.

Q5. Best: t is the first element, e.g. $L = [t, \dots]$ — $\Theta(1)$. Worst: t absent (or last), e.g. a list not containing t — $\Theta(n)$.

Q6a. $O(|V| + |E|)$ — each depot is visited at most once ($|V|$ visits); across the entire run each road is examined at most twice, once from each endpoint ($2|E|$ edge-touches).

Q6b. Complete graph: $|E| = \frac{|V|(|V|-1)}{2} = \Theta(|V|^2)$, so $O(|V| + |E|) = O(|V| + |V|^2) = O(|V|^2)$.